Test Plan Documentation

Interactive House – SG2 Units

Revision History

Date	Version	Description	Author
15/Mar/26	1.0	Initial Test Plan for React Native/Expo Using Jest (1. General and 2. Mobile)	Daniel Jönsson & Ali Daoud
15/Apr/26	1.1	Add UI/Unit Tests to Mobile App and Coverage. Add Table of Contents and New Figures (1, 2, 3)	Daniel Jönsson
8/May/26	1.2	Add Web Testing Section, Test Examples, Coverage Results. Add Update to all Mobile Test Figures. Went from 178 tests to 248 tests. Figures (2, 3, 4, 5, 6, 7, 8) Updated or new	Patrick Nordahl & Daniel Jönsson

Test Plan

1.1 General

In our role as the Units subgroup, we are responsible for the mobile app and web interface that communicates with the server and database. We will test the data exchange between our unit and the database to ensure commands are sent accurately. In line with SDG 3 and SDG 10, we will test the interface to ensure it supports users with high grades of disabilities, emphasizing self-control and independence. The tests are also made to save time, so we do not have to debug everything ourselves.

1.2 Mobile

1.2.1 Scope

Our testing focuses on ensuring that our React Native application can successfully: Receive and render dynamic User Interfaces (UIs) distributed by the server and database from the observed device states in the Arduino Smart Home, such as data from the temperature sensor or control house devices, such as toggling the light "ON" or "OFF" through the database. We will also verify that the UI adapts correctly to the mobile phone's touch screen or alternative inputs like speech recognition.

1.2.2 Testing Approach & Tools

We will use the “Jest Framework” to perform both Unit and UI Testing on individual React Native components and logic functions. We will also use the Expo Go environment to manually validate that the touch-screen interactions (buttons, gestures) work as intended on mobile hardware.

All tests run locally using: “npm test”

The test setup is configured using Jest with Expo support. The following development dependencies are used for testing: jest, jest-expo, @testing-library/react-native,
react-test-renderer, @types/jest.

jest.setup.ts is used for environment setup and required mocks. Test files located in the __tests__ directory. The testing environment is isolated from production configuration files.

1.2.3 Verification & Validation (V&V)

For the verification part, we will use the Jest UI/Unit Test Framework to verify that our internal logic meets the requirements defined in our Requirement Document.

For the validation part, we will assess if the final mobile interface successfully supports users with high grades of disabilities.

1.2.4 Test Examples

Authentication and Access Control
Because our project focuses on a smart home control center context, ensuring only authorized users can access the system is a primary requirement.

We test that the signInWithEmailAndPassword function correctly identifies valid users and provides specific error codes for invalid attempts.

To support the anytime and everywhere nature of ubiquitous computing, we implemented a withTimeout test. This verifies that the app provides immediate feedback if the house server is unreachable, maintaining the user's sense of self-control and independence.

We verify onAuthStateChanged to ensure the app accurately tracks whether a user is currently logged in.

Database Observation and Device Control
A major purpose of this project is to study how to communicate with and observe house devices. Our Firestore tests in mobile validate this interaction.

We use Jest here to mock onSnapshot listeners. This ensures that the mobile unit correctly reflects the status of devices as they change in the database, allowing the user to observe the house remotely.

We also test the control aspect of the project by verifying that toggling a switch in our React Native UI sends the correct update command like turning a light "off" to the database. For the error handling, we test how the app reacts to missing device

configurations or database permission errors.

1.3 Web

1.3.1 Scope

Our web testing focuses on verifying that the Next.js web interface can load the Firebase configuration correctly and support authentication-related user interactions in the browser environment. The tests mainly cover login, signup, Firebase authentication, and Firestore user document creation developed within the SG2 Units subgroup.

The web interface is also tested to ensure that users are redirected to the smart house hub after successful authentication.

1.3.2 Testing Approach & Tools

We use the Jest Framework together with React Testing Library to perform Unit and UI Testing for the web application.

All tests run locally using: “npm test”

The tests are executed in isolated environments using mocked Firebase services to avoid communication with the production database during testing.

The following areas are currently tested:

- **Firebase configuration exports**
- **Login validation and authentication**
- **Signup validation and user creation**

Firestore document creation
User redirection after successful authentication
Friendly error handling for Firebase authentication errors

Coverage reports are also generated to evaluate tested statements, branches, functions, and lines within the web application.

1.3.3 Verification & Validation

For the verification part, we use Jest Unit and UI tests to verify that the authentication flow and Firebase configuration behave according to the requirements defined in our Requirement Documentation.

For the validation part, we manually validate that users can successfully log in, create accounts, and navigate to the smart house hub through the web interface.

Since the SG2 Units subgroup is mainly responsible for the client-side web and mobile interface, the implemented web tests focus on frontend authentication and Firebase-related functionality.

More extensive system-level and integration testing is documented separately in the SG4 subgroup Test Plan documentation.

1.3.4 Test Examples

Firebase Configuration

The Firebase configuration tests verify that the application correctly exports valid Firebase authentication, Firestore, and application instances.

Firebase modules are mocked during testing to isolate the test environment

from the production Firebase configuration.

Authentication and Access Control

The login tests verify that:

- **Validation errors are shown when required fields are missing**
- **Users can successfully sign in using Firebase Authentication**
- **Users are redirected to the /hub page after successful login**
- **Authentication session cookies are created**
- **Firebase authentication errors are translated into user-friendly messages**

The signup tests verify that:

- **Password mismatch validation works correctly**
- **Firebase accounts can be created successfully**
- **User profile information is updated correctly**
- **Firestore user documents are stored successfully**
- **Users are redirected to the /hub page after successful signup**
- **Firebase signup errors are displayed as readable messages**

Figures

Mobile

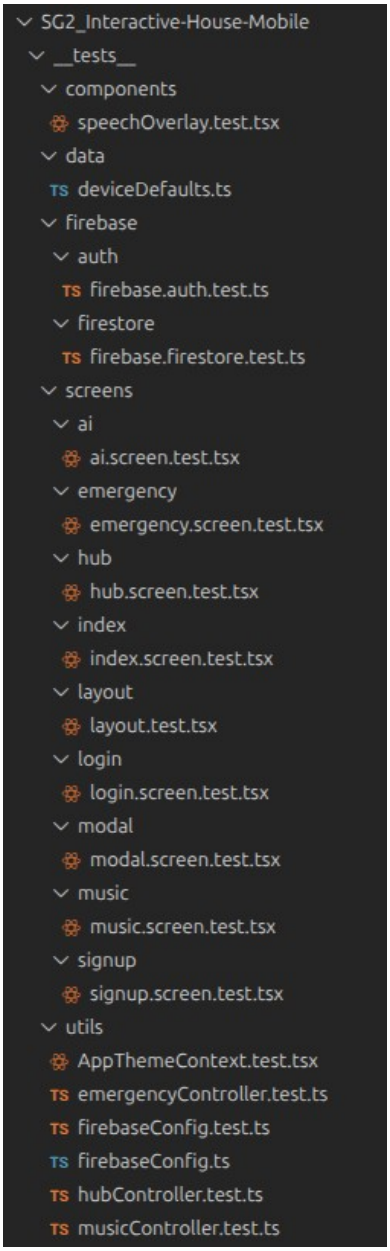


Figure 1: Mobile Testing Structure in Visual Studio Code

```
> interactive-house-mobile@1.0.0 test
> jest

 PASS  __tests__/utils/musicAudio.web.test.ts
 PASS  __tests__/components/speechOverlay.test.tsx
 PASS  __tests__/screens/index/index.screen.test.tsx
 PASS  __tests__/screens/modal/modal.screen.test.tsx
 PASS  __tests__/utils/musicAudio.helpers.test.ts
 PASS  __tests__/utils/AppThemeContext.test.tsx
 PASS  __tests__/firebase/auth/firebase.auth.test.ts
 PASS  __tests__/utils/GuestContext.test.tsx
 PASS  __tests__/utils/musicAudio.native.test.ts
 PASS  __tests__/screens/layout/layout.test.tsx
 PASS  __tests__/screens/emergency/emergency.screen.test.tsx
 PASS  __tests__/utils/hubController.test.ts
 PASS  __tests__/utils/musicAudio.test.ts
 PASS  __tests__/utils/musicController.test.ts
 PASS  __tests__/firebase/firestore/firebase.firestore.test.ts
 PASS  __tests__/utils/emergencyController.test.ts
 PASS  __tests__/data/deviceDefaults.test.ts
 PASS  __tests__/utils/firebaseConfig.test.ts
 PASS  __tests__/screens/ai/ai.screen.test.tsx
 PASS  __tests__/screens/login/login.screen.test.tsx
 PASS  __tests__/screens/signup/signup.screen.test.tsx
 PASS  __tests__/screens/hub/hub.screen.test.tsx (5.573 s)
 PASS  __tests__/screens/music/music.screen.test.tsx (6.314 s)

Test Suites: 23 passed, 23 total
Tests:       248 passed, 248 total
Snapshots:   0 total
Time:        6.939 s
Ran all test suites.
```

Figure 2: Mobile Total Tests

All files

89.62% Statements 1270/1417 **81.68%** Branches 807/988 **87.5%** Functions 245/280 **90.97%** Lines 1219/1340

Press *n* or *j* to go to the next uncovered block, *b*, *p* or *k* for the previous block.

Filter:

File ▲		Statements		Branches		Functions		Lines	
app		92.56%	112/121	90.36%	75/83	90%	36/40	92.24%	107/116
app/(auth)		93.43%	128/137	85.71%	78/91	87.5%	28/32	96.15%	125/130
app/(tabs)		85.75%	668/779	77.95%	495/635	86.92%	133/153	87.48%	643/735
components		93.33%	84/90	91.3%	84/92	100%	8/8	94.04%	79/84
data		100%	1/1	100%	0/0	100%	0/0	100%	1/1
utils		95.84%	277/289	86.2%	75/87	85.1%	40/47	96.35%	264/274

Figure 3: Mobile Coverage

Statements are the individual instructions that make a computer perform an action. A single line of code might contain one statement, or it could contain several.

Branches refer to the decision points in the code anywhere the control flow can branch off in different directions. This usually involves if/else statements, switch cases, or ternary operators (? :).

Functions is the simplest metric. It tracks whether each function or method defined in the code was called at least once during the tests.

Lines measure the physical lines of code in your source file that were executed.

File		Statements		Branches		Functions		Lines	
ai.tsx		89.74%	70/78	87.5%	49/56	92.3%	12/13	93.05%	67/72
hub.tsx		86.63%	214/247	80%	200/250	90.19%	46/51	89.13%	205/230
music.tsx		84.58%	384/454	74.77%	246/329	84.26%	75/89	85.68%	371/433

Figure 4: Mobile Coverage Ai, Hub, Music (Main tabs)

Web

```
$ npm test -- --coverage

> smart-home@0.1.0 test
> jest --coverage

 PASS  __tests__/screens/login/login.screen.test.tsx
 PASS  __tests__/firebase/auth/firebase.auth.test.ts
 PASS  __tests__/screens/music/music.screen.test.tsx
 PASS  __tests__/screens/signup/signup.screen.test.tsx
```

Figure 5: Test Coverage

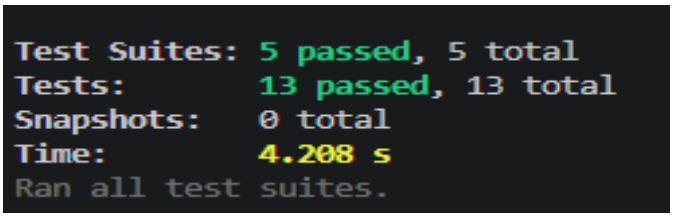
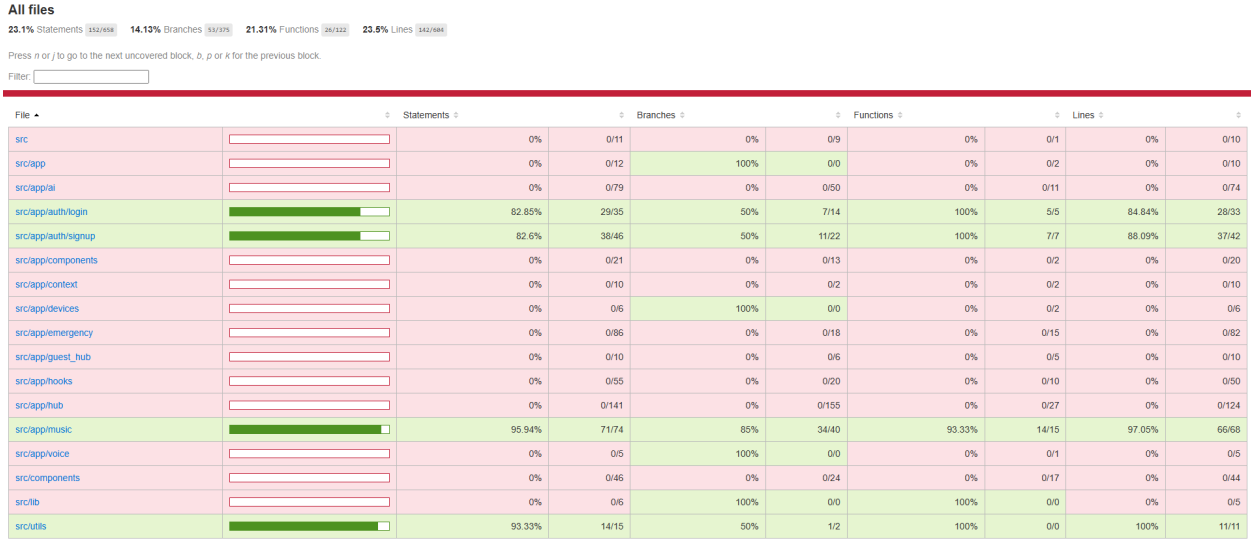


Figure 6: Web Total Test (SG2_Subgroup)



All files

23.1% Statements 152/658 **14.13%** Branches 53/375 **21.31%** Functions 26/122 **23.5%** Lines 142/604

Press *n* or *j* to go to the next uncovered block, *b*, *p* or *k* for the previous block.

Filter:

File		Statements		Branches		Functions		Lines	
src		0%	0/11	0%	0/9	0%	0/1	0%	0/10
src/app		0%	0/12	100%	0/0	0%	0/2	0%	0/10
src/app/ai		0%	0/79	0%	0/50	0%	0/11	0%	0/74
src/app/auth/login		82.85%	29/35	50%	7/14	100%	5/5	84.84%	28/33
src/app/auth/signup		82.6%	38/46	50%	11/22	100%	7/7	88.09%	37/42
src/app/components		0%	0/21	0%	0/13	0%	0/2	0%	0/20
src/app/context		0%	0/10	0%	0/2	0%	0/2	0%	0/10
src/app/devices		0%	0/6	100%	0/0	0%	0/2	0%	0/6
src/app/emergency		0%	0/86	0%	0/18	0%	0/15	0%	0/82
src/app/guest_hub		0%	0/10	0%	0/6	0%	0/5	0%	0/10
src/app/hooks		0%	0/55	0%	0/20	0%	0/10	0%	0/50
src/app/hub		0%	0/141	0%	0/155	0%	0/27	0%	0/124
src/app/music		95.94%	71/74	85%	34/40	93.33%	14/15	97.05%	66/68
src/app/voice		0%	0/5	100%	0/0	0%	0/1	0%	0/5
src/components		0%	0/46	0%	0/24	0%	0/17	0%	0/44
src/lib		0%	0/6	100%	0/0	100%	0/0	0%	0/5
src/utils		93.33%	14/15	50%	1/2	100%	0/0	100%	11/11

Figure 7: Web Coverage (SG2_Subgroup)